Light

GCSE Computer Science - Legal, Ethical and Environmental Issues (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the main purpose of the Data Protection Act 2018?

- ○ **A.** To regulate how personal data is collected, stored, and used by organisations
- ○ **B.** To prevent all data from being stored on computers
- ○ **C.** To make all computer data publicly available
- ○ **D.** To ban the use of encryption

Q2 (1 mark)

Which activity would be illegal under the Computer Misuse Act 1990?

- ○ **A.** Creating a backup of your own files
- ○ **B.** Accessing someone else's computer without their permission
- ○ **C.** Using licensed software correctly
- ○ **D.** Browsing public websites

Q3 (1 mark)

What does the Copyright Designs and Patents Act 1988 protect?

- ○ **A.** Original creative works such as software, music, images, and written content from being copied without permission
- ○ **B.** Physical computer hardware from damage
- ○ **C.** Computer passwords from being guessed
- ○ **D.** Internet connections from being slow

Q4 (1 mark)

What is the key feature of open source software?

- ○ **A.** It is always more expensive than proprietary software
- ○ **B.** It can only be used on one computer
- ○ **C.** The source code is freely available and can be modified by anyone
- ○ **D.** It must be purchased from a single company

Q5 (1 mark)

What is a characteristic of proprietary software?

- ○ **A.** The source code is freely available to everyone
- ○ **B.** Users can modify the code however they want
- ○ **C.** The source code is kept secret and users cannot modify it
- ○ **D.** It is always free to use

Q6 (1 mark)

What is e-waste?

- ○ **A.** Email messages that are no longer needed
- ○ **B.** Electricity wasted by leaving devices on standby
- ○ **C.** Data stored on old hard drives
- ○ **D.** Discarded electronic devices and equipment that can harm the environment if not disposed of properly

Q7 (1 mark)

What is the digital divide?

- ○ **A.** The gap between those who have access to digital technology and those who do not
- ○ **B.** A method of splitting computer files

- ○ **C.** A type of computer virus
- ○ **D.** The process of converting analogue to digital

Q8 (1 mark)

Why do websites ask users to accept cookies?

- ○ **A.** Cookies make websites load faster
- ○ **B.** Privacy laws require websites to inform users about data collection and obtain consent for non-essential tracking
- ○ **C.** Cookies are required by the Computer Misuse Act
- ○ **D.** Websites cannot function without cookies

Q9 (1 mark)

How has the internet contributed to cultural globalisation?

- ○ **A.** By making all websites look identical
- ○ **B.** By preventing people from learning about other cultures
- ○ **C.** By enabling instant sharing of ideas, entertainment, and communication across international borders
- ○ **D.** By only allowing English-language content

Q10 (1 mark)

Why are data centres considered an environmental concern?

- ○ **A.** They consume large amounts of electricity for computing and cooling, contributing to carbon emissions
- ○ **B.** They take up too much office space
- ○ **C.** They are always located in protected forests
- ○ **D.** They produce loud noise pollution only

Q11 (1 mark)

What ethical concern is raised by increasing automation and artificial intelligence?

- ○ **A.** Computers will become conscious and take over
- ○ **B.** All AI systems are already illegal
- ○ **C.** Automation makes products more expensive
- ○ **D.** Job displacement as machines replace human workers in many industries

Q12 (1 mark)

Under the Data Protection Act 2018, what right do individuals have regarding their personal data?

- ○ **A.** The right to demand any company's trade secrets
- ○ **B.** The right to access other people's personal data
- ○ **C.** The right to access, correct, and request deletion of their personal data held by organisations

- ○ **D.** The right to sell other people's data

Q13 (2 marks)

The Computer Misuse Act 1990 defines three main offences. A student guesses their teacher's password and logs into the school system to view exam questions.
Which offence has been committed?

- ○ **A.** No offence - students can access school systems
- ○ **B.** Section 1: Unauthorised access to computer material - accessing a system without permission
- ○ **C.** Only a school rule has been broken, not a law
- ○ **D.** This is legal if no data was changed

Q14 (2 marks)

A student downloads a paid video game from an illegal file-sharing website without paying for it.
Which law has been broken and why?

- ○ **A.** The Copyright Designs and Patents Act - the game is protected creative work, and downloading without a licence is copyright infringement
- ○ **B.** The Data Protection Act - the game contains personal data
- ○ **C.** No law has been broken - digital content cannot be stolen
- ○ **D.** Only the website owner has broken the law

Q15 (2 marks)

A school is choosing between open source and proprietary software for its computer labs.
What is an advantage of choosing open source software?

- ○ **A.** Open source software always has better technical support
- ○ **B.** Open source software is always more secure because it is tested by the company
- ○ **C.** Proprietary software cannot run on school computers
- ○ **D.** No licensing costs, ability to customise the code to the school's needs, and no vendor lock-in

Q16 (2 marks)

A company wants to reduce the environmental impact of its IT systems.
Which action would be most effective?

- ○ **A.** Extending device lifespans, recycling e-waste responsibly, and using energy-efficient equipment
- ○ **B.** Buying new computers every year to get the latest models
- ○ **C.** Printing all documents instead of storing them digitally
- ○ **D.** Leaving all computers running 24/7 for convenience

Q17 (2 marks)

A company monitors all employee emails and internet usage without informing staff.
What privacy and legal issues does this raise?

- ○ **A.** No issues - employers can monitor anything on company equipment
- ○ **B.** Monitoring is only illegal if employees complain
- ○ **C.** Employees have reasonable privacy expectations; monitoring without notice may breach data protection laws and employment regulations
- ○ **D.** Email monitoring is impossible to implement technically

Q18 (2 marks)

A job application system uses AI to screen candidates. The AI was trained on data from previous hiring decisions, which had gender bias. What ethical issue does this demonstrate?

- ○ **A.** AI systems always make fair decisions
- ○ **B.** Algorithmic bias - AI systems can perpetuate and amplify existing discrimination if trained on biased data
- ○ **C.** This only affects the AI company, not job applicants
- ○ **D.** AI bias is not an ethical concern

Q19 (2 marks)

Remote working technology allows companies to hire employees from different countries to work in virtual teams.
What cultural challenges might this create?

- ○ **A.** No challenges - technology removes all barriers
- ○ **B.** Remote workers cannot use the same software
- ○ **C.** International hiring is illegal in most countries

○ **D.** Time zone differences, language barriers, different working practices, and potential misunderstandings due to cultural communication styles

Q20 (2 marks)

A company collects customer email addresses for order confirmations and then sells them to marketing companies without consent.
Which Data Protection Act principle has been violated?

○ **A.** Purpose limitation - data collected for one purpose cannot be used for incompatible purposes without consent

○ **B.** No principle violated - companies own customer data

○ **C.** Data can be used for any purpose once collected

○ **D.** Only government organisations must follow data protection principles

Q21 (2 marks)

A hacker gains unauthorised access to a hospital computer system and deletes patient records.
Under the Computer Misuse Act, which offences have been committed?

○ **A.** Only Section 1 - unauthorised access

○ **B.** No offence - medical records are not protected

○ **C.** Only a civil matter, not a criminal offence

○ **D.** Section 1 (unauthorised access) AND Section 2 or 3 (unauthorised modification of data or intent to impair operation)

Q22 (2 marks)

A software company wants to encourage community contributions while still generating revenue from enterprise customers.
Which licensing approach would best suit this goal?

- ○ **A.** Fully proprietary with no open source version
- ○ **B.** Making everything free with no revenue model
- ○ **C.** Open core model - basic version is open source, but premium features require paid proprietary licences
- ○ **D.** Banning all external contributions

Q23 (2 marks)

Social media platforms collect extensive data about users' behaviour, preferences, and connections to target advertising.
What privacy concern does this raise?

- ○ **A.** No concern - users agreed to terms of service
- ○ **B.** Data collection improves user experience without any drawbacks
- ○ **C.** Users may not fully understand the extent of data collection; detailed profiles enable manipulation and raise questions about informed consent
- ○ **D.** Social media cannot technically collect user data

Q24 (3 marks)

An employee copies confidential company files to a USB drive, accesses their colleague's email account using a guessed password, and downloads pirated accounting software onto the company computer.
Analyse which laws have potentially been broken.

- ○ **A.** Only company policies have been broken, not laws
- ○ **B.** Computer Misuse Act (unauthorised access to colleague's email), Copyright Act (pirated software), and potentially Data Protection Act (mishandling confidential data) - multiple laws apply to different aspects
- ○ **C.** Only the Computer Misuse Act applies to computer-related activities
- ○ **D.** Copyright law only applies to music and films, not software

Q25 (3 marks)

Analyse the environmental impact of a smartphone throughout its complete lifecycle: manufacturing, use, and disposal.

- ○ **A.** Environmental impact only occurs during disposal as e-waste
- ○ **B.** Smartphones have no environmental impact because they are small devices
- ○ **C.** Only the packaging of smartphones affects the environment
- ○ **D.** Manufacturing extracts rare earth minerals with environmental damage; use consumes electricity from potentially fossil fuel sources; disposal creates toxic e-waste if not recycled - impact occurs throughout the entire lifecycle

Q26 (3 marks)

A healthcare system uses AI to recommend treatments. The AI makes a recommendation that leads to patient harm, but no human doctor reviewed the suggestion before it was implemented.
Analyse the ethical issues regarding responsibility and accountability.

- ○ **A.** The AI is solely responsible and should be prosecuted

○ **B.** Accountability is unclear: AI cannot be held legally responsible; questions arise about who is liable (developers, hospital, operators); using AI without human oversight in critical decisions raises ethical concerns about patient safety and informed consent

○ **C.** No one is responsible because computers made the decision

○ **D.** Only the patient is responsible for accepting AI treatment

Q27 (3 marks)

A startup is deciding between using open source database software and purchasing a proprietary database licence.
Analyse the advantages and disadvantages of each approach for a small company with limited budget but needing reliability.

○ **A.** Proprietary is always better for businesses regardless of circumstances

○ **B.** Open source software is always unreliable and should never be used

○ **C.** There is no difference between open source and proprietary software

○ **D.** Open source: free to use, community support, customisable, but may lack guaranteed support and require technical expertise. Proprietary: reliable paid support, tested thoroughly, documentation, but expensive licences and vendor lock-in. Budget favours open source; support needs may favour proprietary

Q28 (3 marks)

A government digitises public services, requiring citizens to apply for benefits, pay taxes, and access healthcare appointments online.
Analyse the cultural and social implications of this digital transformation.

○ **A.** Digital transformation only has benefits with no drawbacks

○ **B.** Benefits: increased efficiency, 24/7 access, reduced costs, faster processing. Concerns: digital exclusion of elderly and low-income citizens without internet access or digital skills; potential for errors in automated systems; loss of personal human interaction for vulnerable people

○ **C.** Nobody prefers using digital services over paper forms

○ **D.** Governments should never use digital technology

Q29 (3 marks)

A company suffers a data breach exposing customer names, addresses, and payment card details. They delay informing customers for three weeks while investigating.
Analyse the Data Protection Act compliance issues.

○ **A.** There are no requirements about breach notification timing

○ **B.** Companies can take as long as they want to investigate before notifying anyone

○ **C.** Only banks need to report data breaches

○ **D.** The Data Protection Act requires notifying the ICO within 72 hours of becoming aware of qualifying breaches; affected individuals must be informed without undue delay if there is high risk to their rights; three weeks delay likely violates these requirements and could result in significant fines

Q30 (3 marks)

Facial recognition technology is proposed for use in public spaces to identify criminals automatically.
Analyse the ethical considerations balancing public safety against privacy rights.

- ○ **A.** Public safety always outweighs privacy concerns without exception
- ○ **B.** For safety: can identify known criminals, find missing persons, deter crime. Against: mass surveillance of innocent people without consent; potential for misidentification causing wrongful accusations; chilling effect on freedom of assembly; risk of function creep beyond original purpose; disproportionate impact on certain demographic groups
- ○ **C.** Privacy rights always outweigh public safety without exception
- ○ **D.** Facial recognition technology is perfectly accurate and has no ethical concerns

Total: **55** marks • Time: **30 min**

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